
Genotype
(deleted gene)
CGT backbone
context h
$E_g = \gamma_g E_g^0$
$\times\ e^{r \tanh(w^\top h_g)}$
$\gamma_g = 0$ if deleted
k_{cat} table
Yeast9 GEM
S, $[\ell_j, u_j]$, GPR
$\text{cap}_j = \sum_{u \in \text{GPR}(j)} k_{cat,u} E_u$
$v_j = \ell_j + (u_j - \ell_j)\,\sigma(z_j)$
bounds exact, capped by $\pm \text{cap}_j$
no projection onto $Sv = 0$
$\ell = a\, v_{\text{growth}} + b$
$v_{\text{growth}} = v_{r_2111}$
BCE(ℓ, y)